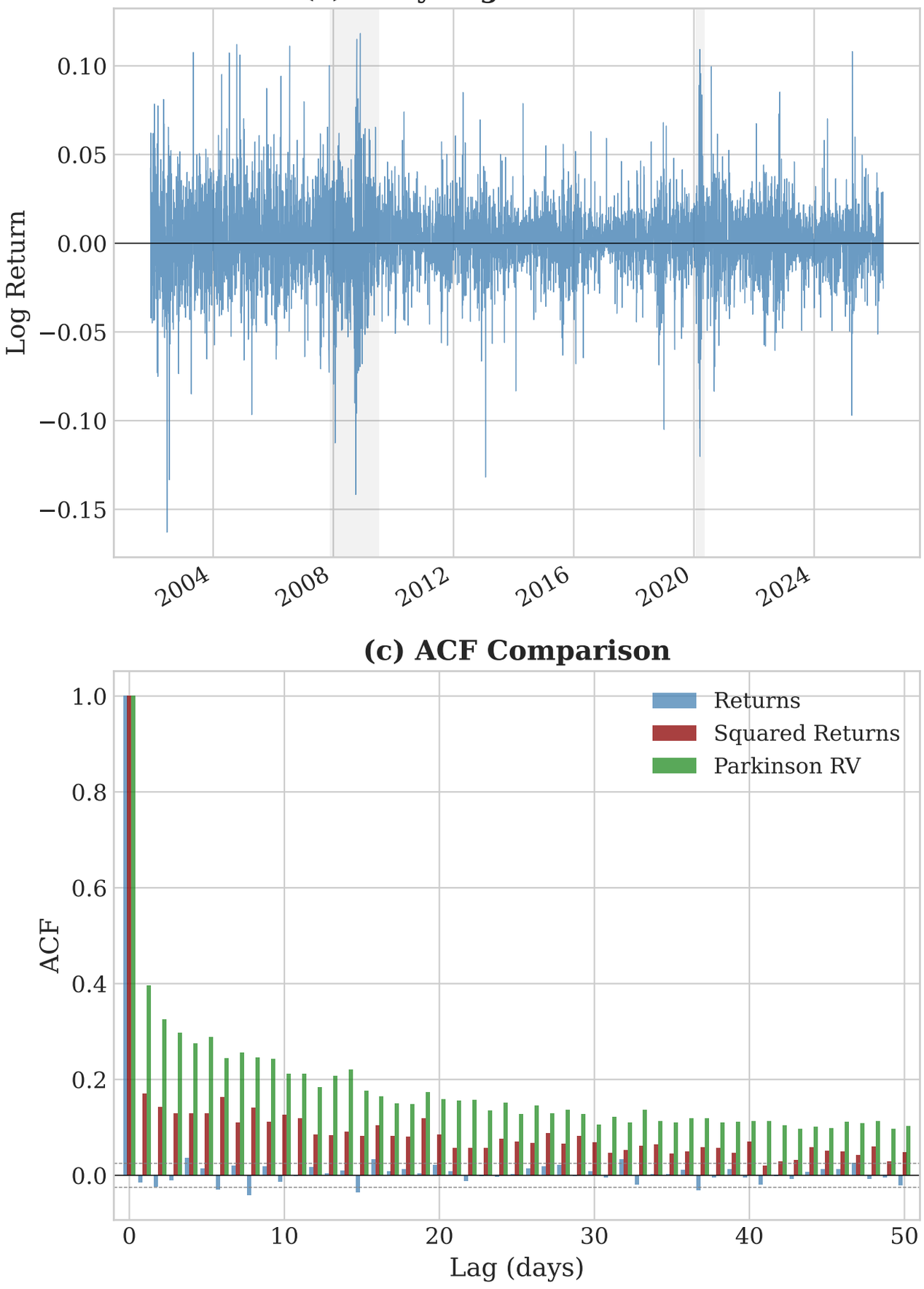
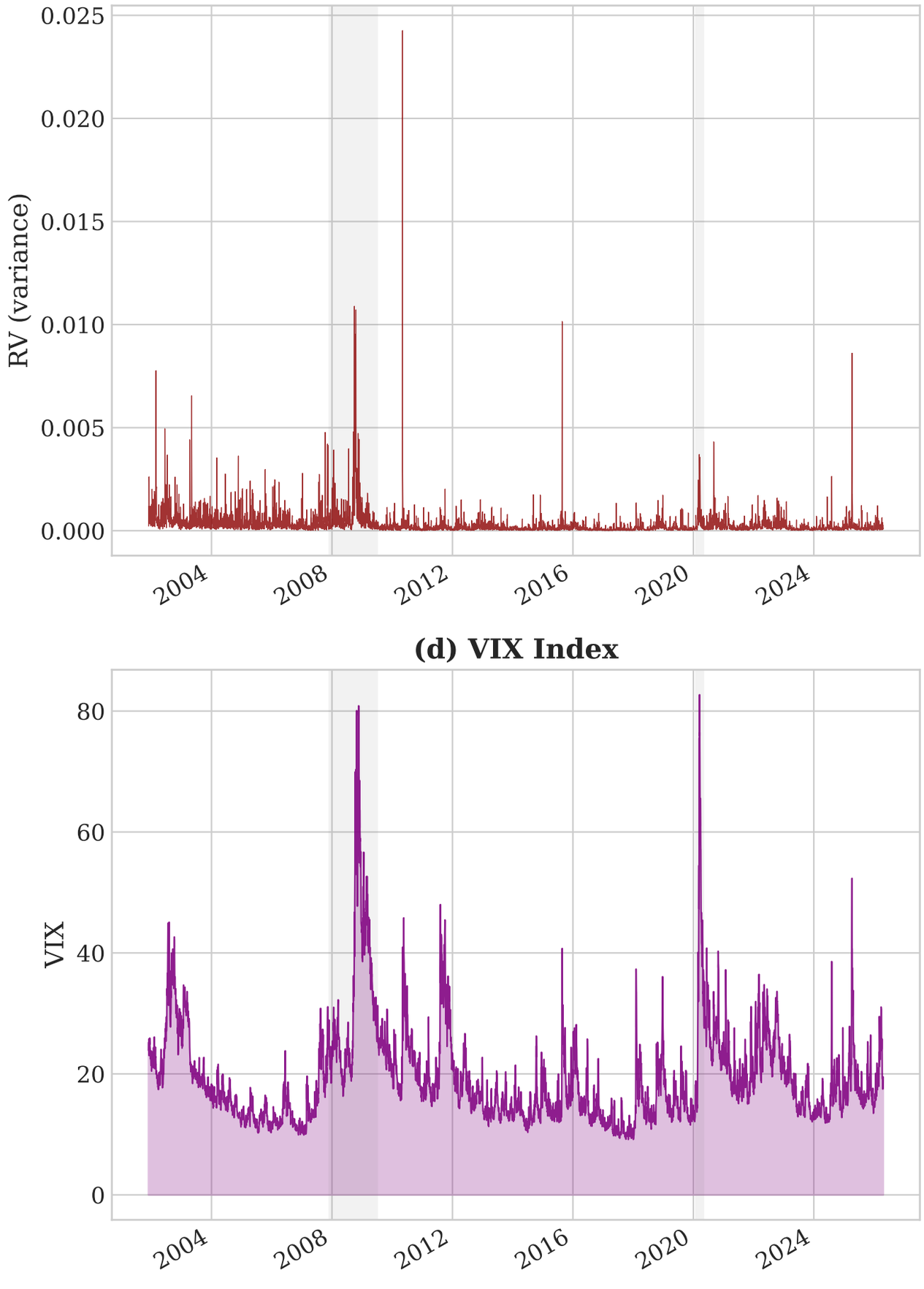

(a) Daily Log Returns: AAPL
Log Return
0.10
0.05
0.00
−0.05
−0.10
−0.15
2004
2008
2012
2016
2020
2024


(b) Parkinson Realized Variance: AAPL
RV (variance)
0.025
0.020
0.015
0.010
0.005
0.000
2004
2008
2012
2016
2020
2024

(c) ACF Comparison
Returns
Squared Returns
Parkinson RV
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
50
Lag (days)

(d) VIX Index
VIX
80
60
40
20
0
2004
2008
2012
2016
2020
2024